

Plan

- Dual of a Primal
- Fundamental theorem of Duality
- The Complementary Slackness Theorem

Dual of a Primal

- Given an LPP (P) (*)
Max $\mathbf{c}^T \mathbf{x}$
subject to $\mathbf{A}_{m \times n} \mathbf{x} \leq \mathbf{b}_{m \times 1}$, $\mathbf{x} \geq \mathbf{0}$

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- The LPP(*) is called a **Primal** problem

- **Theorem 1:** If $\mathbf{x} \in \text{Fea}(P)$ and $\mathbf{y} \in \text{Fea}(D)$, then
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- **Theorem 3:** The **Dual** of the **Dual** (D) (of the Primal (P)) is the **Primal** (P)

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- Clearly (D) **does not** have any **feasible solution**, since the first constraint cannot be satisfied by any non negative u, v

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- Any straight line passing through the origin is a cone
Union of a finite number of straight lines passing through the origin is also a cone
- A **cone** T is said to be a **convex cone** if it is also a **convex subset** of $\mathbb{R}^n$
- **Exercise:** If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ are all vectors in $\mathbb{R}^n$ then check that

$$T = \{\alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + \dots + \alpha_k \mathbf{a}_k : \alpha_i \geq 0 \text{ for all } i = 1, \dots, k\}$$
is a **convex cone**
 $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$ are called **generators** of the **convex cone** T

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- **Corollary 4** implies that if the **Primal** does **not** have a feasible solution then the **Dual** does not have an optimal solution

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- If $\mathbf{x} \in Fea(P)$ and $\mathbf{y} \in Fea(D)$ then $\mathbf{c}^T \mathbf{x} \leq \mathbf{b}^T \mathbf{y}$, hence system (1) has a solution $\Leftrightarrow$ system (1)'' has a solution

$$A_{m \times n} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0} \quad (1)''$$

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- System (1)'' can be written as

$$\begin{bmatrix} A_{m \times n} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{n \times n} & -A_{n \times m}^T \\ -\mathbf{c}_{1 \times n}^T & \mathbf{b}_{1 \times m}^T \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \leq \begin{bmatrix} \mathbf{b} \\ -\mathbf{c} \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \geq \begin{bmatrix} \mathbf{0}_{n \times 1} \\ \mathbf{0}_{m \times 1} \end{bmatrix}$$

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- By **corollary 4**, exactly one of the two systems, (1'') (given above) and (2'') (given below) has a solution.

$$\begin{bmatrix} \mathbf{z}_{1 \times m}^T & \mathbf{w}_{1 \times n}^T & a_{1 \times 1} \end{bmatrix} \begin{bmatrix} A_{m \times n} & \mathbf{0}_{m \times m} \\ \mathbf{0}_{n \times n} & -A_{n \times m}^T \\ -\mathbf{c}_{1 \times n}^T & \mathbf{b}_{1 \times m}^T \end{bmatrix} \geq \begin{bmatrix} \mathbf{0}_{1 \times n} & \mathbf{0}_{1 \times m} \end{bmatrix},$$

$$\begin{bmatrix} \mathbf{z}_{1 \times m}^T & \mathbf{w}_{1 \times n}^T & a_{1 \times 1} \end{bmatrix} \begin{bmatrix} \mathbf{b} \\ -\mathbf{c} \\ 0 \end{bmatrix} < 0 \quad \begin{bmatrix} \mathbf{z}_{1 \times m}^T & \mathbf{w}_{1 \times n}^T & a_{1 \times 1} \end{bmatrix} \geq \mathbf{0}_1$$

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- Check that $[\frac{1}{2}, 3, 0]^T \in \text{Opt}(P)$ and $[1, 1]^T \in \text{Opt}(D)$